RAWABET & PRIVATE CLOUD INFRASTRUCTURE

A SCALABLE CLOUD-NATIVE ECOSYSTEM FOR INTELLIGENT APPLICATION DEPLOYMENT

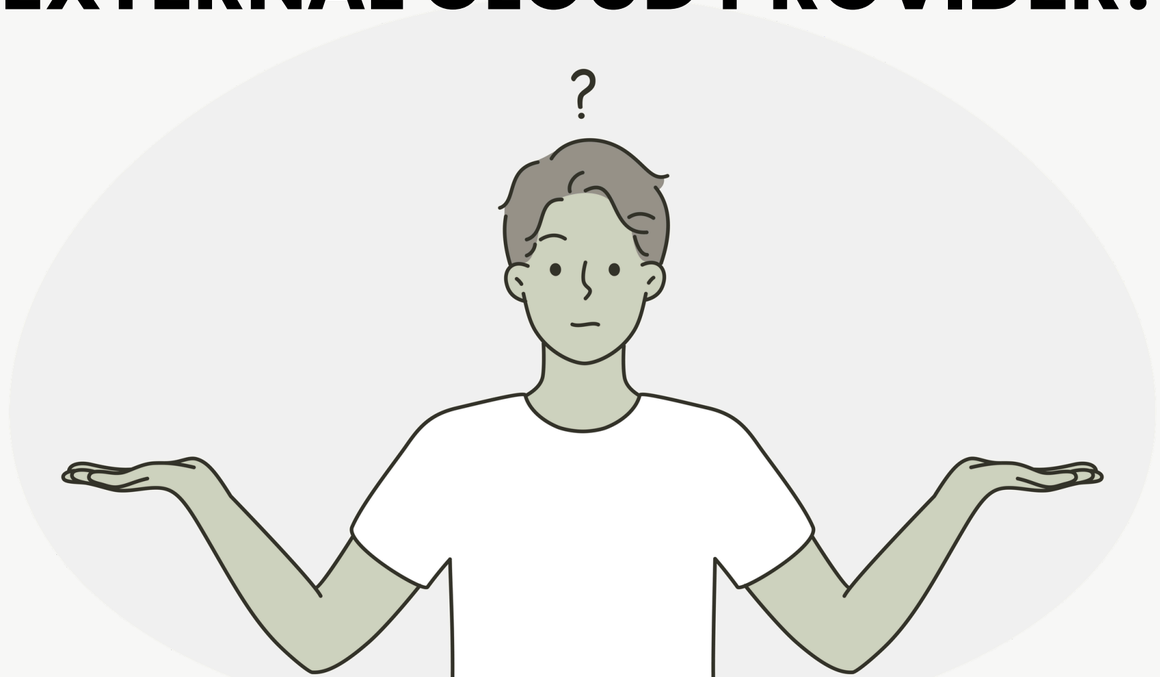
THE PROBLEM WE SET OUT TO SOLVE

- Most modern intelligent applications rely on public cloud providers (AWS, GCP, Azure), creating vendor lock-in, high costs, and limited control over security and data sovereignty.

- Organizations and universities lack affordable, customizable private cloud environments to host and experiment with distributed, scalable applications.



HOW DO WE BUILD AN END-TO-END CLOUD ECOSYSTEM FROM BARE METAL TO INTELLIGENT APPLICATION WITHOUT DEPENDING ON ANY EXTERNAL CLOUD PROVIDER?



WHY BUILD OUR OWN CLOUD?

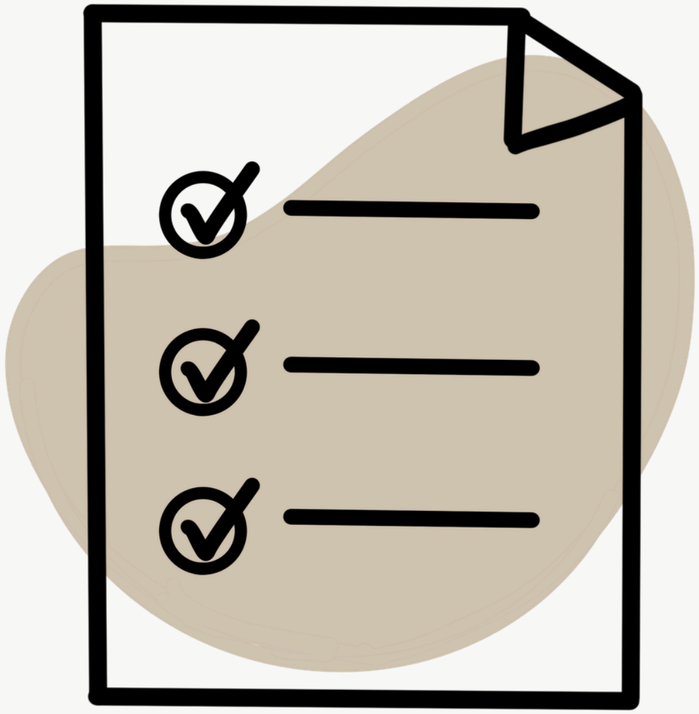
FLEXIBILITY

- Full control over compute, networking, and storage resources via OpenStack.
- On-demand provisioning of virtual machines, networks, and storage volumes.
- Heat orchestration enables infrastructure-as-code: entire environments deployed from templates.

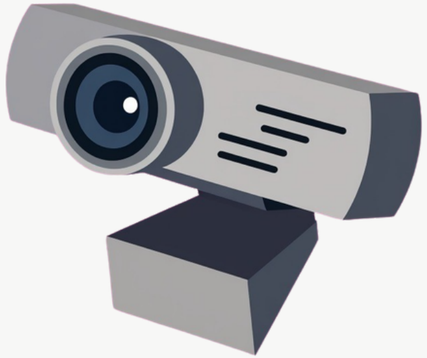
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SCALABILITY

- Horizontal scaling via Nova compute nodes, add capacity without downtime
- Kubernetes auto-scales containerized services based on real workload demand
- Swift object storage handles large-scale unstructured data (media, images, files)



WHY BUILD OUR OWN CLOUD?



SECURITY & SOVEREIGNTY

- Data stays on-premises — no third-party cloud dependency
- Network segmentation and isolation via Neutron
- JWT-based authentication and facial recognition add application-level security layers
- No vendor lock-in; full auditability of the environment



MONITORING

- Prometheus + Grafana provide real-time visibility into infrastructure health, making proactive incident response possible

WHY RAWABET?
WHY NOW?



CONTEXT:

- Tunisia has a rich but underserved cultural scene cinemas, theatres, cultural clubs, and event spaces but most of them still rely on paper, phone calls, or basic spreadsheets to manage their activity.
- People now expect to discover, book, and engage with cultural events online, the same way they use Netflix or social media, but no local platform offers this experience.
- Cultural venues have no single tool to handle bookings, communicate with their audience, and manage their community. They either do it manually or use several disconnected apps.

WHY RAWABET?
WHY NOW?

COMPETITORS:

- Platforms like Eventbrite and Weezevent only handle ticket sales.
- They have no AI, no community space, and no tools for venue managers.

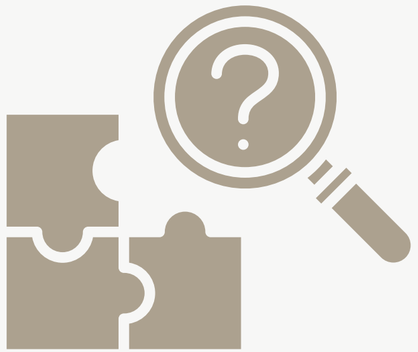


Institutions end up using 3 or 4 different apps to cover what Rawabet does in one place.

WHY RAWABET?
WHY NOW?

THE GAP:

- No platform in the region brings together smart recommendations, scheduling assistance, and community moderation.
- all hosted on a private, secure infrastructure. Rawabet is the first to do this.



WHY RAWABET?
WHY NOW?



WHY IT'S THE RIGHT FIT FOR THIS PROJECT:

Why Rawabet fits the cloud:

- Rawabet is made of several independent services working together, which makes it a great real-world test for our cloud infrastructure.

Why the cloud fits Rawabet:

- Because Rawabet handles live data, AI features, and media content all at once, it needs a powerful and flexible infrastructure, exactly what our private cloud provides.

RAWABET — ONE PLATFORM, INFINITE CULTURE

Module	What it does
	Browse films, showtimes, book seats, receive QR-coded tickets
	Discover and register for cultural events and club activities
	Real-time WebSocket discussions around films and events
	Personalized movie suggestions based on preferences & popularity
	Guides cinema admins in scheduling decisions — analyzes a movie's popularity score, freshness/recency, predicted attendance, and revenue potential before it gets programmed
	AI flags and filters spoilers in live community discussions
	AI generates summaries of long discussion threads
	JWT sessions + facial recognition login
	Digital ticket generation and on-site QR validation

MARKET OPPORTUNITY

500+

Cultural venues in Tunisia
cinemas, theatres, cultural clubs

12M+

Potential end-users
Tunisian population aged 15-50

→Digital tool usage is growing fast, especially after COVID
→No dominant local competitor exists
→Expandable to North Africa & francophone markets

COMPETITIVE LANDSCAPE

Eventbrite / generic ticketing

✘No AI recommendations
✘No community / moderation features
✘Not designed for cultural venues

Local solutions

✘Each feature (tickets, chat, schedule) lives in a separate app
✘No AI, no admin analytics

Rawabet advantage

✓All-in-one: discovery, booking, chat, moderation
✓AI-powered, runs on our own secure cloud
✓Built specifically for cultural venues

REVENUE MODEL

B2B SaaS Subscription

→Monthly fee per cultural venue / institution
→Tiered plans: Basic / Pro / Enterprise

Booking Commission

→Small % per ticket sold through the platform
→Scales with venue activity

Infrastructure Licensing

→Other institutions (universities, public centers) can use our cloud setup.
→High-value, replicable model beyond Tunisia

WHAT MAKES RAWABET DIFFERENT?

FOR USERS:

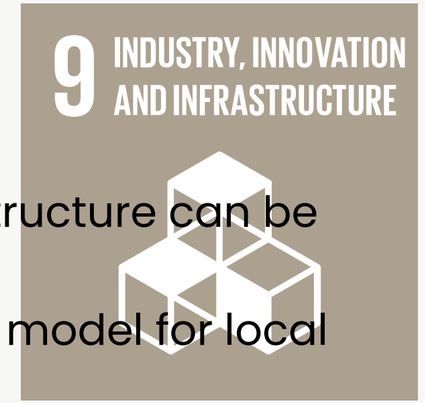
- Personalized cultural discovery — no more scrolling through irrelevant content
- Seamless end-to-end experience: browse → book → attend → discuss, all in one place
- Safe, moderated community spaces (spoiler detection, AI moderation)
- Fast and secure access with facial recognition — no passwords to forget

FOR CULTURAL INSTITUTIONS:

- Centralized management of venues, events, reservations, and audiences
- Real-time data and analytics on attendance and engagement
- Reduced operational overhead through automation
- AI-assisted programming decisions — the platform tells admins which films to schedule and when, based on real popularity signals and revenue forecasts, reducing the risk of empty screenings and maximizing venue profitability
- Moves cinema management from reactive scheduling to predictive, intelligent programming

FOR THE ECOSYSTEM:

- Built on a private cloud — data sovereignty is preserved
- Microservice architecture means each feature can scale independently
- AI features add intelligence without compromising performance



SDG 9 — INDUSTRY, INNOVATION & INFRASTRUCTURE

- The entire project demonstrates that scalable, intelligent digital infrastructure can be built independently, without commercial cloud dependency.
- OpenStack + Kubernetes + DevOps automation represent a replicable model for local digital infrastructure innovation.

RAWABET & THE GLOBAL GOALS

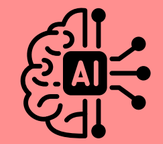


SDG 11 — SUSTAINABLE CITIES & COMMUNITIES

- Cultural centers are pillars of vibrant, inclusive communities. A platform that makes cultural life more accessible strengthens urban social fabric.
- Efficient digital management reduces physical waste (paper tickets replaced by QR codes, printed schedules replaced by real-time digital listings).

BUILT ON A SOLID TECHNICAL FOUNDATION

Application layer
Rawabet



AI engine


Live chat


QR codes


Face ID


Recommendations



Booking

DevOps & Automation
CI/CD & Monitoring




GitLab CI/CD


Prometheus


Grafana

Containerisation & Orchestration
Docker & Kubernetes


Docker


Kubernetes

Infrastructure
Openstack


Neutron


Nova


Swift


Cinder

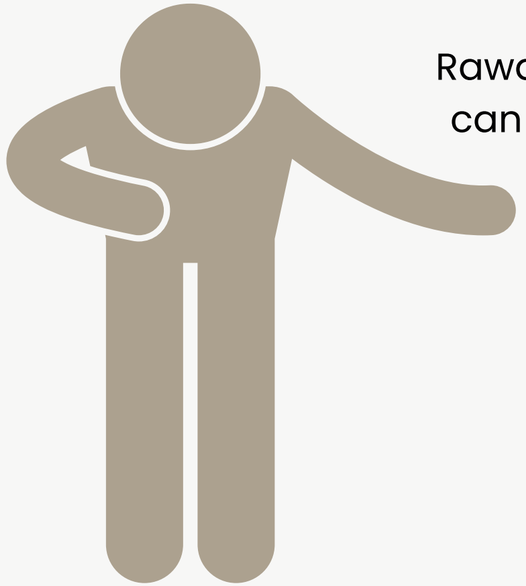

Heat


Horizon


KeyStone


Glance

FROM BARE METAL TO INTELLIGENCE



We didn't just build an app.
We built the cloud it runs on.
Rawabet is proof that a fully sovereign, scalable, AI-powered ecosystem can be engineered from scratch — with the right architecture, the right tools, and the right team.

THANK YOU